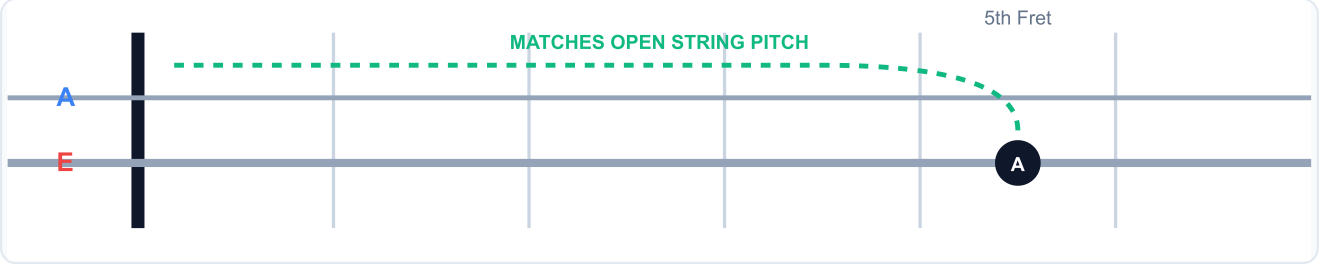
Intuitive Guitar Theory

UNDERSTANDING THE MECHANICS OF MUSIC

1. The 12 Tones & Tuning by Ear

Western music is built on 12 distinct notes before the pitch sequence repeats itself at a higher register (an octave). On a guitar, moving exactly one fret up increases the pitch by one of these steps (a semitone). The strings themselves are tuned to specific intervals to allow the human hand to reach chords comfortably.



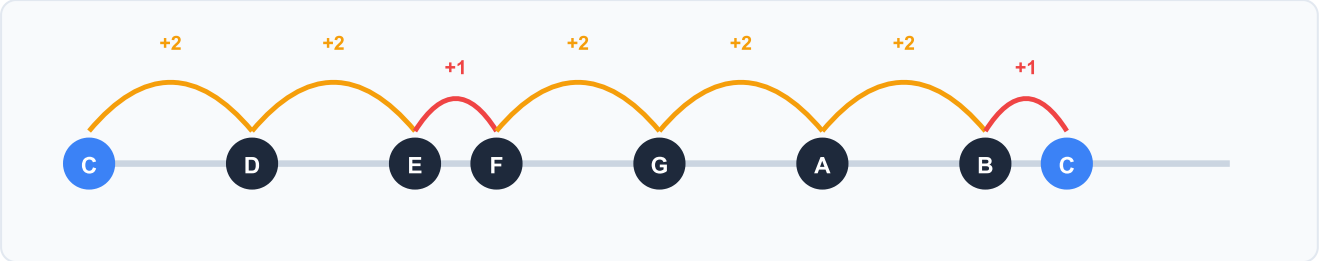
Because of this mathematical layout, you can tune the guitar to itself without an electronic tuner. The standard tuning of a guitar (E-A-D-G-B-E) is designed in **Perfect Fourths**. This means if you start on the Low E string and walk up exactly 5 frets, you arrive at the note 'A'. This 'A' on the 5th fret is physically identical in pitch to the open 'A' string right below it.

You can repeat this process all the way down the neck: touching the 5th fret of the A string gives you 'D' (matching the open D string), and touching the 5th fret of the D string gives you 'G' (matching the open G string).

The One Exception: The jump between the G string and the B string is intentionally tuned slightly closer together (a Major Third) to make full barre chords physically possible for human fingers. Therefore, to tune the B string, you must press the **4th fret** of the G string. To tune the final high E string, you return to the 5th fret of the B string.

2. Deriving the Major Scale

Instead of rote memorizing scale shapes, we must understand the DNA of a scale. A scale is just a curated filter of the 12 chromatic tones. The Major scale is universally built by taking a root note and applying a specific sequence of physical jumps across the frets: **Whole, Whole, Half, Whole, Whole, Whole, Half**.



On a guitar, a "Whole" step means shifting up 2 frets, and a "Half" step means shifting up 1 fret. Notice how the naturally occurring half steps in the musical alphabet (E to F, and B to C) align perfectly with the +1 jumps in our formula. This mathematical alignment is why C Major has zero sharps or flats.

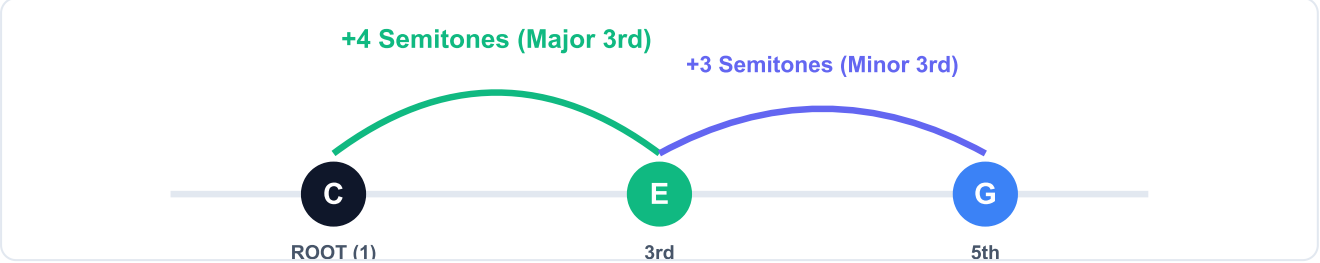
If you apply this exact same sequence starting on a different root note (like E), the formula forces you to introduce sharps to maintain the correct spacing, yielding E - F# - G# - A - B - C# - D# - E.

> **The Relative Minor Formula:** Every major scale has a twin minor scale that shares the exact same collection of notes, just starting from a different emotional center. To find it, shift the root note down exactly 3 frets (3 semitones).
>
> `Relative Minor = Root − 3 Frets`
> For C Major, dropping down 3 frets lands on A. Therefore, A Minor uses the identical notes as C Major (A-B-C-D-E-F-G).

3. The +4 / +3 Triad Code

People often rote learn chord shapes without knowing why they sound the way they do. A chord is not arbitrary; it is built by stacking musical intervals—specifically "Thirds." A standard triad consists of three notes: a Root (1), a Third (3), and a Fifth (5).

There are two types of Thirds used to build chords: A **Major 3rd** is exactly 4 semitones (+4 frets) above its starting note. It sounds bright and happy. A **Minor 3rd** is exactly 3 semitones (+3 frets) above its starting note. It sounds dark and melancholic.

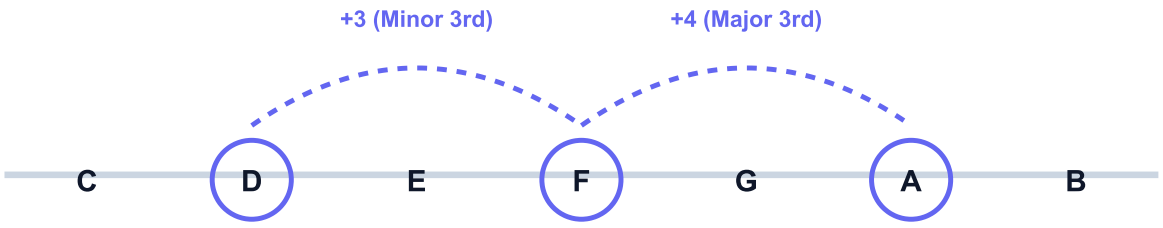


To build a **Major Triad** (like C Major), you start at the root (C), go up +4 frets to find your middle note (E), and from there, go up another +3 frets to find your top note (G). Thus, the genetic code of a Major chord is **+4, then +3**.

Conversely, to build a **Minor Triad**, the formula reverses. You start at the root, go up +3 frets (establishing a sad Minor 3rd immediately), and from there go up +4 frets. The outer boundaries of both chords remain identical—the Perfect 5th provides the strong walls, while the internal 3rd dictates the emotional flavor.

4. Deriving Diatonic Harmony Intuitively

You don't have to guess which chords sound good together in a key; they are mathematically generated by the scale. If we take the C Major scale (C-D-E-F-G-A-B) and build a triad starting from *every single note*, restricting ourselves to only using notes within the scale, a predictable pattern emerges by simply skipping every other note.



Let's test this intuitively starting on the second note, **D**. We take D, skip E, take F, skip G, and take A. Our resulting notes are D - F - A. Let's analyze the intervals: D to F is exactly +3 frets. F to A is exactly +4 frets. Because the formula is +3 then +4, we have naturally derived a **Minor Chord** just by sticking to the scale notes.

If you repeat this "skip one, take one" process for every note, the physics dictate that the I, IV, and V chords will naturally be Major, the ii, iii, and vi chords will naturally be Minor, and the vii chord will be Diminished. You don't have to memorize which is which; you can derive them by counting frets.

5. Mapping the Triad to the Fretboard

Now that we know a C Major chord is explicitly the notes C, E, and G, how do we play it on a 6-string guitar? The intuitive step is to map out every single C, E, and G that exists on the first three frets, and then pick a combination that our hand can physically reach.

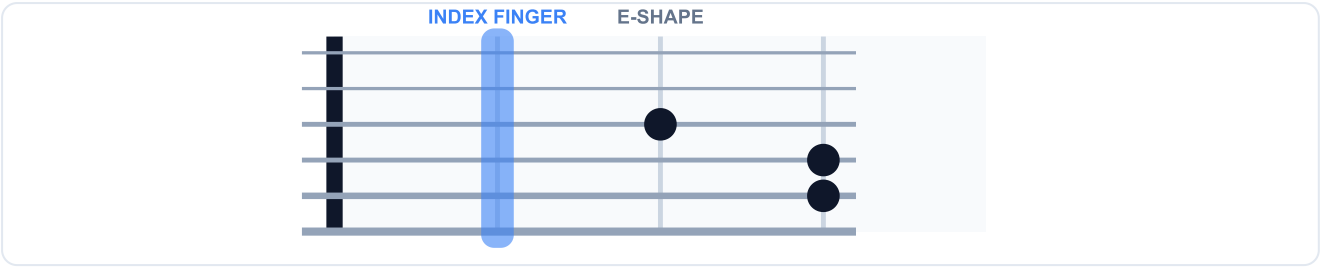


By letting the G and high e strings ring openly, pressing C on the A string, E on the D string, and C again on the B string, we form the classic C Major open shape. We are hitting the triad notes multiple times across different octaves to create a fuller sound.

Why do we mute or skip the low E string? Even though there is technically an 'E' note available on that open string (and E belongs to the C Major chord), adding it as the lowest sounding note completely changes the gravity of the chord. To the human ear, the lowest pitched note acts as the foundation. We want C to be our lowest note so the chord feels grounded and resolved, hence we skip or mute the thickest string.

6. Barre Chords: Shifting the Nut

If we apply our triad math (+4, +3) to the root note F, we find that an F Major chord consists of F, A, and C. If we try to map those notes onto the fretboard near the nut, we quickly realize that none of the open strings naturally provide an F, A, or C without awkward stretching. How do we play it?



The intuitive solution leverages the geometric nature of the guitar. Think about an open E Major chord. The physical guitar "nut" is essentially holding down the 0th fret for you across all 6 strings while you place your fingers on the 1st and 2nd frets. The root note is the open Low E string.

To turn this into an F Major chord, we must shift the entire mathematical structure up by one fret (because E to F is a half-step). We physically slide the E-Major finger shape up one fret, but we now lack the open strings—the nut didn't move. So, we must lay our free index finger completely flat across the 1st fret, forcing it to act as a movable, human "nut" (the Barre). The fretboard calculates the intervals for you.

7. The Step-by-Step Chord Derivation

By relying on physical logic instead of rote memorization, you can now construct any chord on the fretboard from scratch. If you forget a chord, follow this intuitive thought process to rebuild it:

- **Step 1: Identify the Request.** If you need to play a G Minor chord, your starting Root note is G.
- **Step 2: Apply the Internal Math.** Recall the code for a minor triad: Root, +3 semitones, +4 semitones.
- **Step 3: Calculate the Notes.** Start on G. Move up 3 semitones (G -> G# -> A -> Bb). From Bb, move up 4 semitones (Bb -> B -> C -> C# -> D). Your triad notes are exactly G, Bb, D.
- **Step 4: Find Your Anchor.** Locate the root note 'G' on the thickest strings (Low E or A) to serve as your bass note. The lowest G is on the Low E string, 3rd fret.
- **Step 5: Geometric Mapping.** Since you are anchoring on the E string, mentally visualize the movable barre shapes (the E Major and E Minor shapes). Because you want a G *Minor*, you visualize sliding the open E Minor shape up the neck until your index finger acts as the nut behind the 3rd fret G note.

> **Intuition Check:** You don't need to manually map all the Bb and D notes individually across the neck under pressure. The geometry of the guitar ensures that if you slide an open Minor shape up to a G root, all the internal strings are automatically shifting up the exact mathematical intervals required to land perfectly on Bb and D.